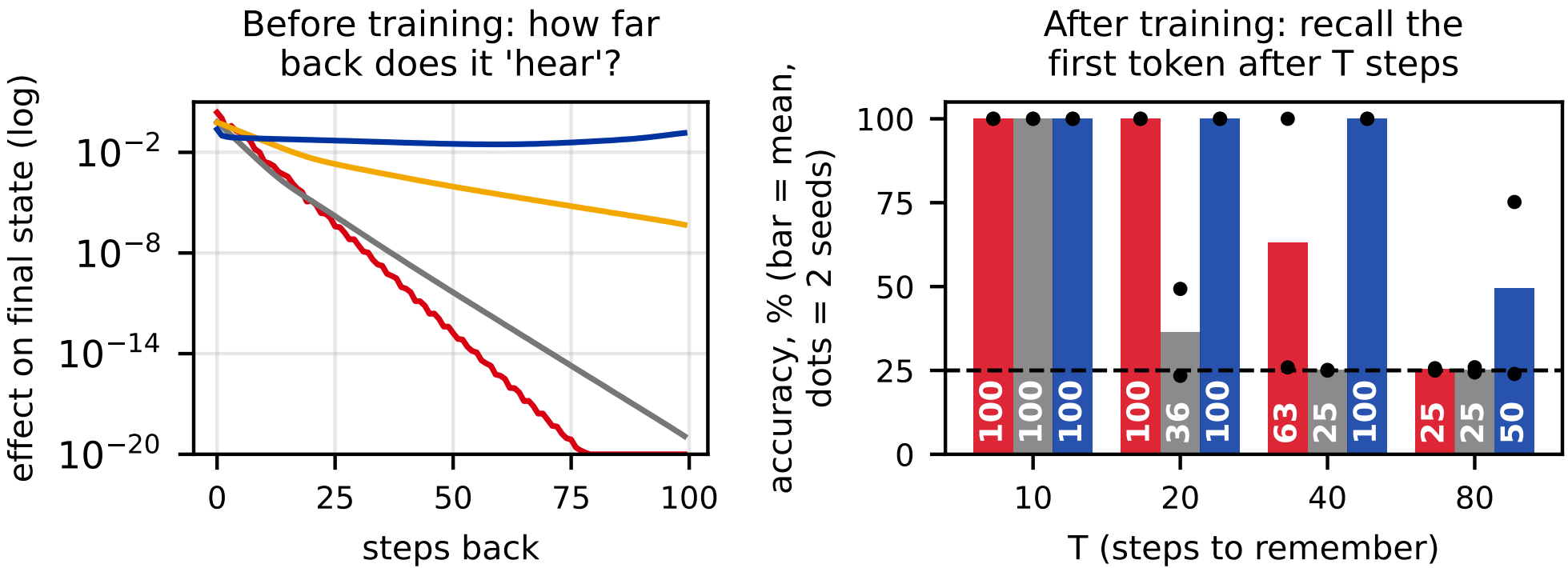
Before training: how far back does it 'hear'?
effect on final state (log)
10^{-2}
10^{-8}
10^{-14}
10^{-20}
0
25
50
75
100
steps back
After training: recall the first token after T steps
accuracy, % (bar = mean, dots = 2 seeds)
100
75
50
25
0
100
100
100
100
36
100
63
25
100
25
25
50
10
20
40
80
T (steps to remember)
vanilla RNN
LSTM, forget bias 0 (PyTorch default)
LSTM, bias 1 (Keras default)
LSTM, bias 3 (gate open)
chance (25%)